

Piyush Sonawane

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LinkedIn | GitHub

EMBEDDED, IoT AND ROBOTICS ENGINEER

Final year Electronics and Telecommunication Engineering student at NBNSCOE with hands on experience building embedded and IoT systems using ESP32, Arduino, sensors, actuators, wireless communication, and real time telemetry. Experienced in sensor interfacing, embedded firmware, safety automation, relay and motor control, Firebase, hardware software integration, and system debugging. Strong foundation in C, C++, Python, microcontrollers, communication protocols, Linux, Git, and web based monitoring systems with a focus on practical cyber physical systems.

TECHNICAL SKILLS

- **Programming:** C, C++, Python, JavaScript, TypeScript, Embedded C, SQL
- **Microcontrollers:** ESP32, Arduino Nano, Arduino IDE, GPIO, PWM, ADC, Timers, Interrupts, Embedded Firmware
- **Communication Protocols:** UART, I2C, SPI, Serial Communication, Bluetooth, WiFi, TCP/IP, HTTP, REST, MQTT, IoT
- **Sensors and Actuators:** Load Cells, HX711, MQ-6, PZEM-004T, MPU6050, HC-SR04, GPS, DHT, OLED, DC Motors, Servo, Relays, Motor Drivers
- **IoT and Cloud:** Firebase RTDB, IoT Telemetry, Remote Monitoring, Device Control, Real Time Data Acquisition, Cloud
- **Robotics and Control:** Motor Control, Sensor Integration, Actuator Control, Safety Automation, Feedback Systems, Embedded Control, Hardware Software Integration
- **Software and Engineering:** React.js, Node.js, REST APIs, Git, GitHub, Docker, Linux, VS Code, KiCad, PCB Prototyping, Debugging, Testing, Calibration
- **Robotics Familiarity:** ROS2 Concepts, Nodes, Topics, Services, TF, Gazebo Concepts, Sensor Fusion, Localization, Kinematics, Control Systems

EXPERIENCE

MCMH

June 2026 to September 2026

Software Development Engineer Intern

- Contributed to application development, backend integration, debugging, testing, and deployment workflows for a real world software product.
- Implemented and improved features using Git based development practices while collaborating in a structured engineering workflow.
- Applied software engineering and DevOps practices to develop maintainable and reliable solutions.

EDUCATION

N. B. Navale Sinhgad College Of Engineering, Solapur

Bachelor of Engineering in ENTC | 8.43 SGPA till Sem VI

Maharashtra, India

2023 to 2027

MGSM's Arts, Science and Commerce College, Chopda

HSC | 77.23%

Maharashtra, India

2022 to 2023

Pankaj Vidyalaya Sec., Chopda

SSC | 94.40%

Maharashtra, India

2020 to 2021

PROJECTS

OneGuard | IoT Gas Cylinder Safety and Monitoring System

- Developed an ESP32 based IoT safety system for real time LPG cylinder weight and gas leakage monitoring using four load cells, HX711, and MQ-6 gas sensor.
- Integrated load sensor combinator, MPU6050, DS3231 RTC, buzzer, relay, servo, and embedded hardware for multi sensor acquisition, actuator control, and safety monitoring.
- Designed sensor data acquisition and IoT telemetry workflows with threshold based safety alerts, embedded debugging, calibration, and hardware software integration.

OneFlux | IoT Energy Monitoring and Safety Automation System

- Built an ESP32 and PZEM-004T based IoT system for real time voltage, current, power, frequency, energy, cost, and CO2 monitoring.
- Implemented firmware safety logic for undervoltage, overvoltage, overcurrent, and overpower conditions with autonomous relay trips on sustained faults.
- Developed a React and Vite dashboard with Firebase RTDB for live telemetry, safety alerts, relay commands, state verification, and remote monitoring.

Gesture Controlled Vehicle | ESP32 Wireless Robotics System

- Built a four wheel robotic vehicle using ESP32, Arduino, HC-05 Bluetooth communication, L298N motor driver, motors.
- Designed a modular control architecture separating wireless command processing, motion logic, and motor control for real time gesture based navigation.

LEADERSHIP AND LANGUAGES

Leadership and Achievements

- Associate General Secretary and Departmental Cultural Secretary at NBNSCOE
- Hackathon Success: 1x Winner, 2x Top 4 Finishes

Languages

- English
- Hindi
- Marathi